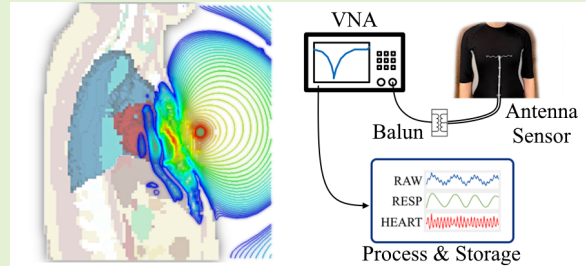


Single Antenna Bio-Sensing for Non-invasive Respiratory and Cardiac Activity Monitoring

Mehran Ahadi, Amine Miled, Marc-André Dugas, and Younès Messaddeq

Abstract—The Single Antenna Bio-Sensing (SABioS) method is investigated for non-invasive respiratory and cardiac activity monitoring. Using an antenna sensor installed over the chest of a subject, the vital signs can be captured by analyzing changes in the antenna's reflection coefficient due to two contributors, being the variations in the dielectric composition of the body and the structural deformations of the antenna due to thoracic expansion during the respiratory cycle. The operation principle of this method is elaborated and validated through simulations, and its agreement with a medical-grade reference device is studied via a preliminary experimental setup involving 14 volunteers. The results were analyzed using Bland-Altman analysis, linear regression, and various error metrics, and the maximum calculated mean absolute errors were 0.06 bpm for breathing rate and 0.14 seconds for inspiration/expiration time, demonstrating a strong agreement with the medical reference device. The paper also briefly explores the potential of SABioS in cardiac monitoring and detecting breathing pauses, as well as its versatility in operating with different antenna types. The SABioS method provides a non-invasive, comfortable, and accurate solution for continuous vital sign monitoring without any dependency to external devices.

Index Terms—Antenna Sensor, Biosensor, Vital Signs, Respiratory Monitoring, Cardiac Monitoring, Wearable Antenna, Non-Invasive, Sleep Apnea



I. INTRODUCTION

RESPIRATION monitoring is crucial for diagnosing diseases, monitoring the progression of various health conditions, and predicting severe events. Studies show that a Breathing Rate (BR) higher than 27 breaths per minute (bpm) is a predictor of cardiac arrest in hospitals [1]. Relative changes in BR show medical instability of the patient [2] and can indicate the need for ICU admission [3]. It is also shown that patients with higher BR have an increased mortality rate in hospitals [4]. Additionally, continuous follow-up on the respiratory function is essential for patients with chronic respiratory conditions such as asthma, sleep apnea, and chronic obstructive pulmonary disease (COPD).

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Despite the critical role of respiratory monitoring in diagnosing diseases, predicting severe events such as cardiac arrest, and assessing the evaluation of chronic respiratory conditions, it is still clinically measured either manually, through cumbersome systems such as a flow meter built into a mouthpiece (Spirometry), a mask, or measurement of end-tidal CO₂, or with wired sensors whose reliability is often disputed by clinicians [5]. In ICU settings, the continuous respiratory monitoring equipment is often limited to less-reliable techniques such as Electrocardiogram (ECG)-derived [6] and Photoplethysmogram (PPG)-derived methods which are highly affected by the artifacts [7], [8] and chemical properties of the skin [9], [10], leading to reduced quality of care, increased patient risks, and a higher workload for medical staff due to the need for frequent manual examinations, particularly due to repeated false alarms. These limitations underscore the need for a portable and non-invasive method for monitoring respiratory activity with an accuracy comparable to medical standard devices.

Recently, various non-invasive respiratory monitoring methods have been published in the literature. Examples include a flexible optical fiber for cardiorespiratory monitoring [11], the Waveflex optical fiber sensor [12], and a flexible fiber with Bragg gratings (FBG) [13]. Fiber-based methods usually require complex and power hungry light sources and detector circuits, which can limit their integration potential [14]. However, they are advantageous in environments where electrical or magnetic interference is a concern, such as during Magnetic

Resonance Imaging (MRI). Strain-based methods, such as Piezoresistive elements [15], Respiratory Inductance Plethysmography (RIP) [16], and capacitive stretch sensors [17], offer another approach to respiratory monitoring. These typically require tight chest belts, with the more comfortable versions embedded in textile at the expense of accuracy. Additionally, facemasks equipped with humidity and temperature sensors provide yet another approach for monitoring respiration [18].

Radiofrequency-based methods for respiratory monitoring also exist, such as radar-based techniques [19], which generally rely on the Doppler shift effect. However, radar systems require patients to remain relatively static and within a limited coverage range, restricting mobility. RFID-based methods [20] and Bluetooth RSSI-based detection [21] also use radiofrequency, but these approaches lack true mobility as they depend on the proximity of an external device. Another promising technique is Near-field Coherent Sensing (NCS) [22], which analyzes the transmission signal between two closely spaced antennas placed on the body.

In this work, the Single Antenna Bio-Sensing (SABioS) method is investigated for non-invasive detection of vital signs through analyzing the reflection coefficient of an antenna sensor placed on the subject's body. We show the operation principle and capability of SABioS in measuring respiratory and cardiac activity of a subject using a single antenna sensor. Section II discusses the operation principle of this method and the sensor specifications for an optimal detection. A preliminary experimental setup is presented in Section III, with the results presented in Section IV. Finally, Section V discusses the agreement with the medical standards, the comparison with the state-of-the-art, the capabilities, and the challenges.

II. OPERATION PRINCIPLE OF SENSOR SYSTEM

A. Respiratory Cycle Detection Mechanism

The respiratory cycle comprises inspiratory and expiratory phases, mechanically facilitated by the intercostal muscles and diaphragm. During diaphragmatic contraction, the diaphragm pulls on the lungs and pushes the abdominal organs downward, causing the stomach to rise outward. This action also causes intra-thoracic pressure to drop, expanding the lungs and triggering the inspiration process. In contrast, during expiration, the diaphragm and intercostal muscles relax, pressure is removed from the abdominal organs, lung volume is reduced, and air is expelled [23]. The changes in the circumference of the body's torso during the respiration process are linked with the tidal volume and, therefore, can be an indicator of different stages of the respiration cycle [24].

In addition to chest expansion, the electromagnetic specifications of the human body also change near the thorax area throughout the respiration cycle. According to Gabriel et al., at 915 MHz, while deflated human lungs exhibit a relative permittivity of $\epsilon_r \approx 51.4$, an inflated lung at peak inspiration is filled with air ($\epsilon_r = 1$), which causes the effective value of the lungs' relative permittivity to fall to $\epsilon_r \approx 22$ [25], [26]. This drastic change in the dielectric properties of the thorax along with the movements of internal organs can be another indicator reflecting the respiration cycle.

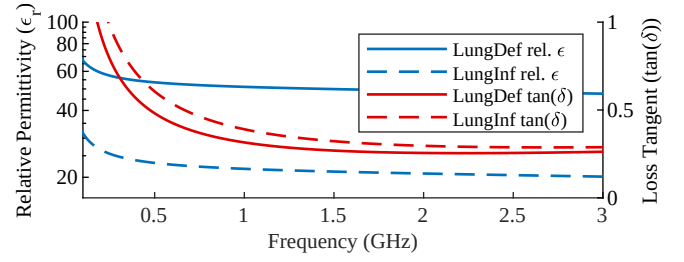


Fig. 1. Relative Permittivity (ϵ_r) and Loss Tangent ($\tan(\delta)$) of human lungs in inflated (LungInf) and deflated (LungDef) states

A single antenna can be utilized as a sensor to intercept respiratory activity through the changes in the dielectric composition of the subject's body during the respiration cycle. To detect the dielectric changes, the antenna should be placed close to the thorax area, where the internal organ movements and the changes in dielectric permittivity of the body occurs. This configuration can be considered similar to an antenna placed over a substrate with a time-variable ϵ_r throughout the respiration cycle of the person. Alterations in the effective dielectric permittivity of the body leads to a shift in the antenna's input impedance [27], [28]. This phenomenon will be explained in detail in the following. In the context of this paper, this effect will be referred to as the Permittivity Shift Effect (PSE).

In the case of employing stretchable or flexible materials for fabrication, the antenna can also function as a stretch- or deformation-sensitive sensor where geometrical changes induce a shift in its input impedance [29]. This method can be leveraged as a second contributor influencing the antenna's impedance, allowing the antenna sensor to pick up the thoracic expansion due to respiration. Additionally, such flexible stretch-sensitive antenna sensor can also be positioned over the abdomen region to monitor respiratory activity via outward abdominal movements due to diaphragm contractions. This contributor will also be referred to as the Antenna Deformation Effect (ADE) in this paper.

A shift in the input impedance of the antenna sensor, whether due to PSE or ADE, can be detected by measuring its reflection coefficient, commonly represented by the S-parameter S_{11} , which is defined as (1), where Z_{ant} is the antenna sensor's input impedance, and Z_0 is the system's characteristic impedance [30].

$$S_{11} = \frac{Z_{ant} - Z_0}{Z_{ant} + Z_0} \quad (1)$$

The alterations in S_{11} can represent the changes in the dielectric composition close to the antenna sensor, and also the structural deformations of a stretchable antenna sensor, which can correlate with the vital signs of the subject.

B. Impact of Subject's Body on Antenna Sensor

The tissue dielectric properties employed in this study are derived from [25], proposing a four Cole-Cole model to compute the dielectric permittivity of tissues, accounting for the diverse electromagnetic properties of various biomolecules within the tissue. The human body is a composition of dense

and lossy dielectric materials. From the surface of the chest to the lungs, there are layers of different tissues including skin, fat, bones, and muscles, with the latter being the densest with $\epsilon_r \approx 55$ and a loss tangent of $\tan(\delta) = 0.34$ at 915 MHz. The dielectric specifications of the lungs in inflated (max inspiration) and deflated (max expiration) states are also shown in Fig. 1.

In a lossy dielectric medium where $\tan(\delta) > 0$, the electric field $\mathbf{E}$ at the distance r can be written as:

$$\mathbf{E}(r) = \mathbf{E}_0 e^{-\gamma r} \quad (2)$$

where $\mathbf{E}_0$ is the field at the boundary of the lossy dielectric and free space, and γ denotes the propagation constant. The γ is a complex value, as shown in (3), where ω is the angular frequency, j is imaginary unit, ϵ is dielectric permittivity, μ is dielectric permeability, α is the attenuation constant, and β is the phase constant [30].

$$\gamma = \alpha + j\beta = jk = j\omega\sqrt{\mu\epsilon} \quad (3)$$

Therefore, (2) can also be written as:

$$\mathbf{E}(r) = \mathbf{E}_0 e^{-\alpha r} e^{-j\beta r} \quad (4)$$

where $e^{-\alpha r}$ is commonly known as the attenuation term. The α and β can also be acquired as (5) and (6), where ϵ_r is the relative permittivity of the lossy medium, ϵ_0 is free space permittivity, and μ_0 is free space permeability [30].

$$\alpha = -\omega \operatorname{Im}[\sqrt{\mu\epsilon}] = -\omega\sqrt{\mu_0\epsilon_0\epsilon_r} \operatorname{Im}[\sqrt{1 - j\tan(\delta)}] \quad (5)$$

$$\beta = \omega \operatorname{Re}[\sqrt{\mu\epsilon}] = \omega\sqrt{\mu_0\epsilon_0\epsilon_r} \operatorname{Re}[\sqrt{1 - j\tan(\delta)}] \quad (6)$$

The distribution pattern of electric and magnetic fields of an antenna in free space depends solely on the antenna's design and materials. However, introducing a dielectric material near the antenna disrupts its near-fields pattern. Dielectrics, when exposed to an electric field, become polarized. This polarization separates their internal charges slightly, creating tiny dipoles that generate their own electric field. The interplay between the fields of the antenna and these induced dipoles leads to a redistribution of the overall field pattern. Consequently, the electric and magnetic fields generated by the antenna in the presence of a nearby dielectric differ from the fields it would generate in free space. Due to Ampère's circuital law, this altered field distribution necessitates a readjustment of the antenna's current distribution, affecting its effective electrical length and consequently, its input impedance [27].

The shift in the impedance will depend on the distance between the antenna and the dielectric, as well as the relative permittivity and loss tangent of the dielectric composition. To illustrate the trend, a straight half-wave dipole antenna is simulated using Finite Element Method (FEM) once in free space and once close to a dielectric as presented in Fig. 2. The PSE is evident in the rise of the current's imaginary component, implying a modification on effective electrical length and impedance of the antenna.

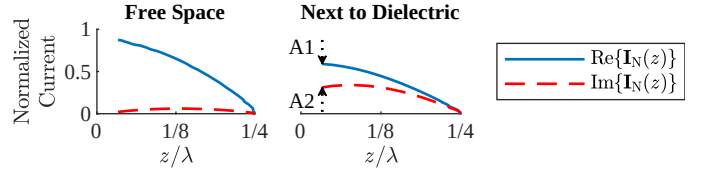


Fig. 2. Current distribution over an arm of a half-wave dipole antenna in free space vs. adjacent to a dielectric block with $\epsilon_r = 4$. The $\mathbf{I}_N(z)$ denotes the current at point z over the antenna, normalized to the maximum source current $\mathbf{I}_0$. Feed point is located at zero and λ represents wavelength. Black arrows annotated as A1 and A2 indicate the shift in real and imaginary parts of $\mathbf{I}_N(z)$ at the feeding point due to the dielectric presence compared to the scenario in free space. Only one dipole arm is shown due to the structure symmetry.

C. Antenna Sensor Design Considerations

The most basic antenna type for SABioS can be a straight, thin, center-fed, dipole antenna. The input impedance Z_{dipole} for such antenna in free space, including both resistive and reactive components, can be calculated through the induced electromotive force method as shown in (7), where $\eta_0 = 120\pi$ is the impedance of free space, $C = 0.577\dots$ is Euler's constant, l is the dipole physical length, and $\operatorname{Ci}(x)$ and $\operatorname{Si}(x)$ are Cosine and Sine Integral functions [31].

$$Z_{dipole} = \frac{\eta_0}{4\pi \sin^2(\frac{\beta l}{2})} (C + \ln(\beta l) - \operatorname{Ci}(\beta l) + j \operatorname{Si}(\beta l)) \quad (7)$$

The input impedance of dipole antenna can be directly affected due to ADE if the antenna is stretched along its length (l). Moreover, the term βl can be rewritten based on electrical length (G), as shown in (8), highlighting the influence of PSE on Z_{dipole} .

$$\beta l = \frac{2\pi}{\lambda} l = 2\pi G \quad (8)$$

The choice of antenna type for SABioS is not restricted to dipoles. All antenna types generally demonstrate strong near-fields [27] allowing for a significant influence on their input impedance due to PSE when placed close to body. Also, antenna characteristics are generally sensitive to geometrical factors, paving the way towards employing stretchable antennas for leveraging ADE. For instance, the impedance of an elliptical loop antenna is influenced by its electrical length, the lengths of its major and minor axes, as well as their ratio [32], [33], demonstrating its sensitivity to dielectric load and also mechanical stretching in two dimensions. Similarly, microstrip patch antennas, as another example, have their impedance directly influenced by the effective permittivity (ϵ_{eff}) based on the dielectric substrate located below the patch, and the subject body placed above the patch. They are also sensitive to patch dimensions along two axes, allowing for sensor applications based on antenna deformation [27].

To benefit the most from ADE, as one of the two sensing contributors, the antenna should be structurally stretchable along a dimension affecting its impedance. It needs to be placed on a wearable, such as a T-shirt, a belt, or a skin patch, and be oriented to have its stretchable axis along the circumference of chest or abdomen, so the thoracic or abdominal expansions during the respiratory cycle can deform the

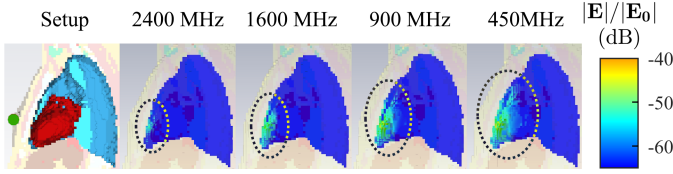


Fig. 3. The Setup tile shows the cross section of the simulated phantom model with dipole antenna installed over the chest, presenting heart as red, lungs as blue, and antenna position as green. Intensity of E-field reaching lungs and heart is shown for different frequencies, normalized to its maximum over the antenna conductors surface.

sensor. The performance will be dependent on the sensitivity of antenna's impedance to the applied strain.

The other contributor, being PSE, will be more significant if the subject's body intersects with the major part of antenna's near-fields. A directive antenna concentrates fields in a specific direction, resulting in a higher gain and more intense fields. Therefore it is best to employ a directive antenna installed as close as possible to the subject's body, oriented to direct its fields pattern towards the subject.

The electric and magnetic fields of an antenna can have several components decaying at a rate of $1/r^n$, where $n = 1, 2, \dots$. The components having $n > 1$ are very dominant in the near-field region of the antenna but decay quickly with distance and lose dominance in the far-field region [27]. This decay weakens interactions with distant dielectrics and minimizes their impact on antenna impedance, effectively limiting the sensing range of the antenna. The near-field region ends at Fraunhofer Distance (R_{fd}) for electrically large antennas, which is the beginning of the far-field region and is defined as $R_{fd} = 2D^2/\lambda$ where D is the largest dimension of the antenna [27]. Rewriting the equation as (9) reveals that choosing a lower frequency, resulting in a larger wavelength (λ), extends the near-field region further. It also shows that using an antenna with a larger electrical length (G) contributes more significantly, with a squared effect.

$$R_{fd} = 2 \frac{D^2}{\lambda} = 2 \left(\frac{D}{\lambda} \right)^2 \lambda = 2G^2 \lambda \quad (9)$$

Another limiting factor for PSE is the lossy nature of body tissues that convert a portion of the electric and magnetic fields power to heat, limiting the reach of antenna fields within the body. The penetration depth ($\delta_p = 1/\alpha$) is the distance at which the field attenuates by $1/e \approx 36.8\%$ due to dielectric losses [27]. Minimizing the attenuation constant (α) increases δ_p and improves the response of PSE, which is possible by adjusting the only tunable parameter in (5), being the frequency. Therefore, selecting a lower frequency not only compensates for field decay by extending the near-field range but also aids in increasing penetration depth by reducing the attenuation.

A straight, thin, dipole antenna is simulated close to a human body phantom model to prove the validity of the discussions above. The antenna is redesigned for various resonance frequencies while placed over the chest of the Hugo phantom model from CST Voxel Family (Dassault Systèmes, Vélizy-Villacoublay, France), maintaining a 2 mm gap from

the skin. The phantom model, with a resolution of 5 mm^3 , incorporates material data sourced from [25]. Results shown in Fig. 3 highlight the tendency of electric fields to penetrate deeper inside the body at lower frequencies. However, integrating antennas operating at very low frequencies into wearables poses a fabrication challenge due to their larger size requirements. Furthermore, as shown in Fig. 1, the $\tan(\delta)$ of tissues tend to increase as frequency drops below 500 MHz, making further reductions of frequency less beneficial in reducing dielectric losses.

Due to the finite radius and penetration depth of the antenna's near-fields, a gap between the antenna and the body surface will weaken the PSE. If the gap distance is electrically short ($\ll \lambda$), the weakening will be minimal, but it worsens as the gap approaches the R_{fd} limit. This weakening can be somewhat mitigated if the gap is filled with a dielectric material denser than air, such as textiles with an ϵ_r ranging from 1.1 to 7.1 [34]. This material reduces the abrupt dielectric discontinuity between air and the body's highly dense tissues, functioning similarly to an anti-reflection coating [35]. As a result, the transition becomes more gradual, slightly improving the coupling of the fields with the tissues. Therefore, the presence of clothes between the antenna and the body primarily weakens the PSE due to the introduced gap, though the dielectric properties of a denser textile can partially offset this effect. This weakening remains minimal unless the operating frequency is high enough that the thickness of the textile becomes comparable to the wavelength.

III. EXPERIMENTAL SETUP

A preliminary experimental setup is prepared as a proof-of-concept to show the feasibility of this method in monitoring the respiratory activity, as shown in Fig. 4. For this study, a sinusoidal dipole antenna made of a flexible conductive polymer is utilized as the sensor, as previously introduced in [29], to benefit from the both PSE and ADE contributors. The conductive polymer used in the antenna fabrication is biocompatible and water-resistant. It has a thickness of 1.4 mm and a resistance of $\approx 8 \Omega \cdot \text{cm}^{-1}$. The material comprises

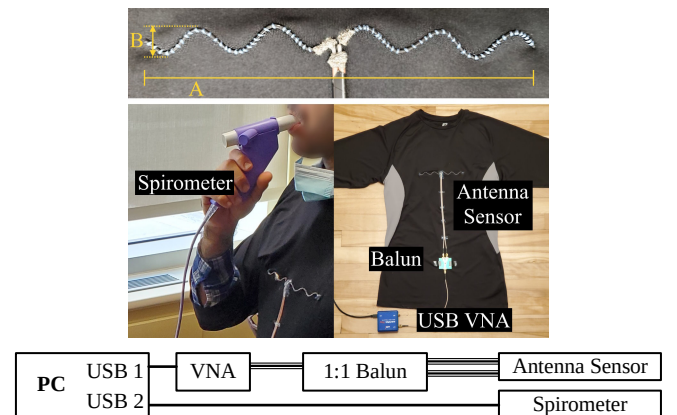


Fig. 4. The sinusoidal antenna sensor, the experimental setup, and the reference medical device. The dimensions of the antenna sensor is $A = 74.54 \text{ mm}$ by $B = 5.65 \text{ mm}$.

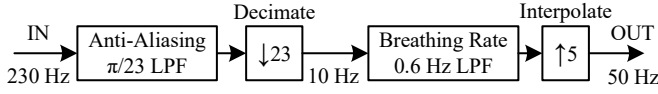


Fig. 5. The multi-rate signal processing pipeline.

41 wt% multi-walled carbon nanotubes (MWCNTs) and 59 wt% poly(ethylene-co-vinyl acetate) (PEVA) polymer [29]. The fabricated sinusoidal antenna resonates at 827 MHz in free space and approximately 808 MHz when placed on the body. It is sewn onto the chest area of a T-shirt and connected to a 1:1 balun to convert the unbalanced signal from the coaxial cable into a balanced feed for the dipole.

The antenna sensor is driven using a USB-powered Vector Network Analyzer (VNA). The VNA is calibrated using Short-Open-Load (SOL) method to remove the effects of cables and connectors from the measured S_{11} parameter. A custom Java program is developed using the publicly-available Application Programming Interface (API) library of the VNA device to measure and record the antenna's S_{11} over the single frequency of 827 MHz for 120 seconds. The signals are recorded with the fixed sampling rate of 230 Hz and logged in a Comma-Separated Values (CSV) file.

Alongside the sensor, a hospital-grade spirometer (Easy on-PC, ndd Medizintechnik AG, Zurich, Switzerland) is also operated. The spirometer signal is recorded as the reference to validate the sensor measurements. Volunteers were asked to wear the T-shirt with the embedded antenna over the chest, and breathe normally through the spirometer during the test. For volunteers with smaller body circumference, safety pins were used to adjust the T-shirt from back, ensuring a reasonably snug fit and preventing excessive drooping in the front section where the sensor was installed. This experimental protocol has been approved by Ethics Review Board of CHU de Québec - Université Laval hospital.

An adult's respiration rate typically falls between 12 and 20 breaths per minute, translating to a frequency range of 0.2 Hz to 0.33 Hz. Given the high and non-configurable sampling rate of the VNA device, the Nyquist frequency is determined to be 115 Hz, in which the ideal cutoff frequency of 0.33 Hz lies at the 0.3% edge of its spectrum. Designing a sharp low-pass filter for this condition is computationally demanding. Therefore a multi-rate signal processing approach is taken according to the diagram shown in Fig. 5.

TABLE I
ANTHROPOMETRIC CHARACTERISTICS OF VOLUNTEERS

Male Volunteers					Female Volunteers				
#	Age	Weight (Kg)	Height (cm)	BMI (Kg/m ²)	#	Age	Weight (Kg)	Height (cm)	BMI (Kg/m ²)
1	29	96	190	26.6	1	30	54	163	20.3
2	42	90	180	27.8	2	31	56	164	20.8
3	58	66	178	20.8	3	25	60	165	22.0
4	31	94	180	29.0	4	26	69	162	26.3
5	28	84	175	27.4	5	28	75	168	26.6
6	40	85	165	31.2					
7	28	90	177	28.7					
8	26	107	186	30.9					
9	34	74	175	24.2					

To analyze the recorded signals, the magnitude of the antenna's reflection coefficient ($|S_{11}|$) is loaded in MATLAB (MathWorks, Natick, MA, US). The signal is decimated by a factor of 23 after applying an anti-aliasing filter, then a 0.6 Hz low-pass filter (LPF) is applied to discard the unwanted noises above 36 bpm, followed by a spline interpolation by a factor of 5 to match with the 50 Hz sampling rate of the reference signal acquired via the spirometer device. All the filters are FIR type, designed using the Parks-McClellan method to avoid phase distortion, which can alter signal waveform. Filters are applied forward and backward directions to achieve zero-phase filtering, preserving the original signal's integrity for accurate analysis.

IV. RESULTS

A group of 9 male and 5 female volunteers participated in this preliminary experiment with their consent. They were asked to wear and operate the sensor while breathing simultaneously through the reference spirometer. Table I summarizes the anthropometric data of volunteers. The volunteers were invited from different sexes, ages, Body Mass Indexes (BMI), and physiques to ensure the inclusivity of the results. Samples of acquired signals are shown in Fig. 6 for male and female volunteers. As presented, the recorded signals show a great correlation with the exchanged air volume measured by the spirometer. However, to show the agreement between two different measurement methods in clinical applications, the Bland-Altman method is often employed [36]. The linear regression is also computed between the two measurement methods to examine the potential presence of a proportional bias in the sensor.

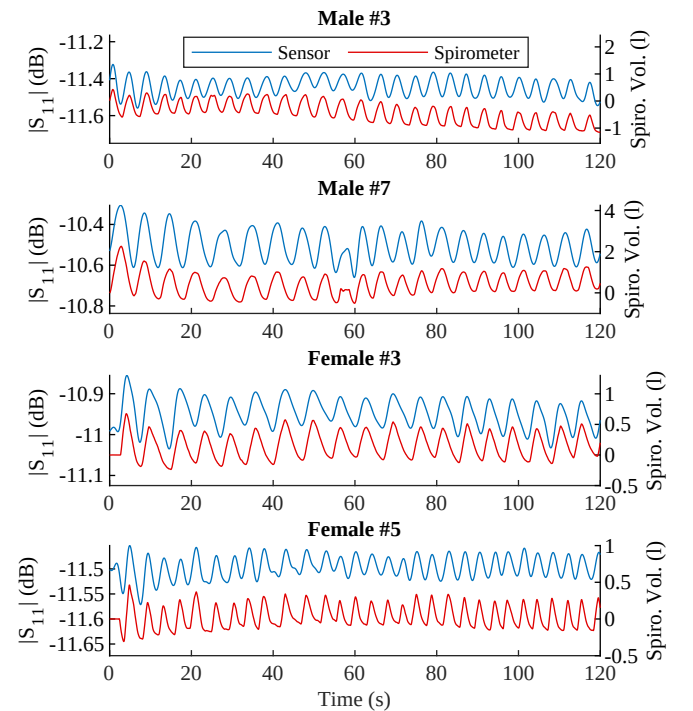


Fig. 6. The sample signals from four of the volunteers. The illustrated signals were chosen from diverse age and BMI ranges of each gender to show the robustness of SABIOS.

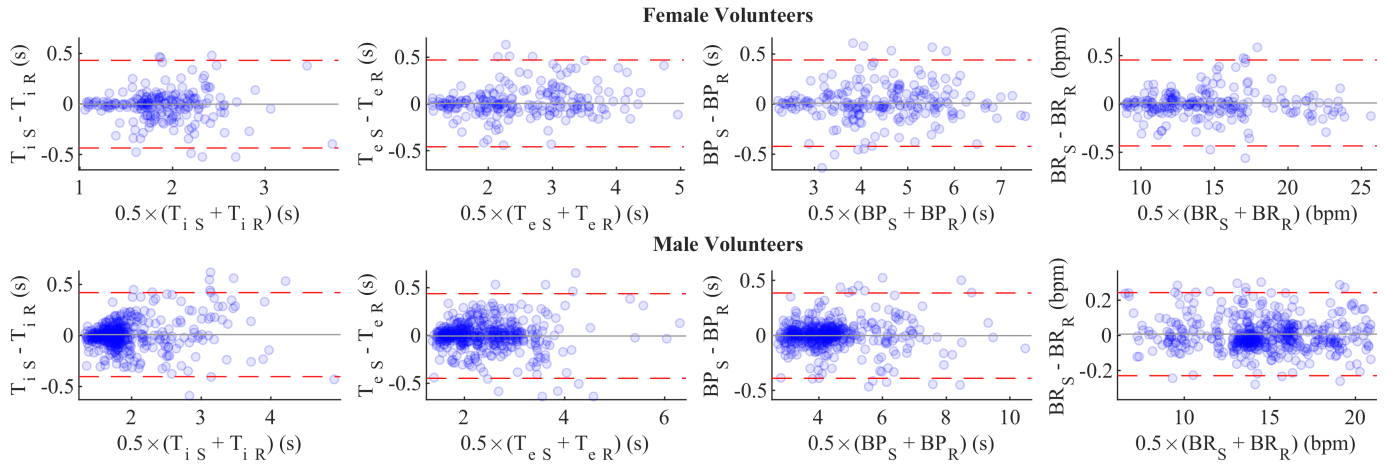


Fig. 7. The Bland-Altman Analysis of volunteers. Gray line shows the MoD, and red dashed lines show the 95% LoA range. The S and R indices of parameters indicate the sensor and the reference, respectively.

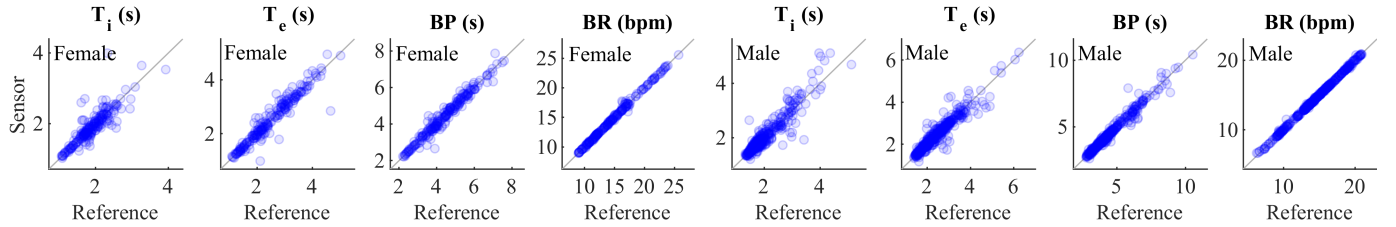


Fig. 8. Scatter Plots of the respiratory parameters as measured by the sensor vs. the reference device for male and female volunteers. The respective respiratory parameter is indicated on top of each figure. The unity slope line is shown in gray.

TABLE II

CALCULATED BLAND-ALTMAN PARAMETERS (MoD $\pm$ LoA)

Parameter	Female Volunteers	Male Volunteers	All Volunteers
T_i (s)	-0.0017 ± 0.433	0.0081 ± 0.412	0.0051 ± 0.418
T_e (s)	0.0063 ± 0.464	-0.0058 ± 0.442	-0.0020 ± 0.449
BP (s)	0.0081 ± 0.431	-0.0018 ± 0.388	0.0012 ± 0.401
BR (bpm)	0.0108 ± 0.446	0.0074 ± 0.237	0.0085 ± 0.316

TABLE III

PROPORTIONAL BIAS (PB) AND LINEAR REGRESSION

Parameter	Female Volunteers			Male Volunteers		
	PB	r	p	PB	r	p
T_i	0.9261	0.8753	< 0.001	1.0330	0.9406	< 0.001
T_e	0.9917	0.9611	< 0.001	0.9450	0.9555	< 0.001
BP	0.9971	0.9828	< 0.001	0.9878	0.9868	< 0.001
BR	1.0006	0.9982	< 0.001	0.9946	0.9993	< 0.001

TABLE IV

CALCULATED ERRORS FOR EACH RESPIRATORY PARAMETER

Parameter	Gender	MAE	MAPE	MedAE	MedAPE	RMSE
T_i (s)	Female	0.12	6.17 %	0.06	3.15 %	0.2206
	Male	0.12	5.39 %	0.06	3.53 %	0.2098
T_e (s)	Female	0.13	5.26 %	0.07	2.83 %	0.2362
	Male	0.13	5.02 %	0.06	2.84 %	0.2253
BP (s)	Female	0.14	3.25 %	0.09	1.94 %	0.2196
	Male	0.10	2.12 %	0.05	1.17 %	0.1976
BR (bpm)	Female	0.12	0.81 %	0.06	0.42 %	0.2272
	Male	0.08	0.58 %	0.05	0.34 %	0.1209

Assuming $AE(i) = |x_r(i) - x_s(i)|$ and $APE(i) = |AE(i)/x_r(i)| \times 100\%$:
 MAE = mean(AE), MAPE = mean(APE), MedAE = median(AE),
 MedAPE = median(APE), RMSE = sqrt(mean(AE²)).

The Bland-Altman method creates a scatter plot with y-axis showing the difference between the two readings, and the x-axis showing their average. The mean of difference (MoD) reveals the systematic bias, the spread of the datapoints reflects agreement between the methods, and the limits of agreement (LoA) are $\pm 1.96 \times$ standard deviation, representing the 95% percentile confidence range. To calculate the parameters of Bland-Altman plot, the peaks and valleys of the filtered signals are detected, and Inspiration Time (T_i), Expiration Time (T_e), and Breathing Period (BP) are calculated for each respiratory cycle in both sensor and reference signals. Additionally, Breathing Rate (BR) is also estimated based on a moving average of BP for 5 respiratory cycles. The recorded cycles are analyzed, calculating the difference and average for each respiratory parameter from both the sensor and the reference. The resulting plots are shown in Fig. 7. for male and female volunteers, with parameters summarized in Table II.

The scatter plots in Fig. 8 illustrate the linear relationship between the sensors. For each respiratory parameter, the proportional bias is determined by calculating the slope of a linear regression between the two measurement methods. The proportional bias, along with the Pearson's correlation coefficient (r) and the p -value, are presented in Table III. Additionally, the mean absolute error (MAE) and mean absolute percentage error (MAPE) are computed for each observation using the sensor (x_s) and the reference (x_r) data points. The less outlier-sensitive metrics, median absolute error (MedAE) and median absolute percentage error (MedAPE), are also calculated. The root mean square error (RMSE) is provided as well. Table IV

TABLE V
SUMMARY OF SIMILAR WORKS

Paper	Parameter	Reported Metrics
NCS [22]	BR (bpm)	$r = 0.93$, $MoD = 0.05$, $LoA = \pm 5.75$
Flexible Optical Fiber [11]	BR (bpm)	$MAPE = 5.25\%$, $RMSE = 1.28$, $MoD = -0.2250$, $LoA = \pm 0.8275$
WaveFlex Fiber Sensor [12]	BR (bpm)	$MoD = 0.14$, $LoA = \pm 2.13$, $r = \sqrt{0.91} = 0.95$
Flexible FBG [13]	BR (bpm)	$MAPE = 1.6\%$
Piezoresistive [15]	BR (bpm)	$MoD = 0.05$, $LoA = \pm 1.97$
RIP [16]	BR (bpm)	$r = 0.84$, $p < 0.01$
Temperature Facemask ^a [18]	BR (bpm)	$MAE = 0.69$, $MAPE = 1.64\%$, $MoD = -0.02$, $LoA = \pm 2.37$
Radar [19]	BR (bpm)	$MAE = 0.95$, $MoD = -0.85$, $LoA = \pm 2.18$
RFID ^b [20]	BR (bpm)	$100\% - MAPE = 94.21\%$
Bluetooth RSSI [21]	T_i (s)	$MoD = -0.019$, $LoA = \pm 0.511$
	T_e (s)	$MoD = 0.032$, $LoA = \pm 0.568$
	BP (s)	$MoD = 0.003$, $LoA = \pm 0.490$
	BR (bpm)	$MoD = 0.011$, $LoA = \pm 2.137$

^a High intensity interval training ^b Over the chest, reader at 1.2 m

summarizes these calculated error values.

V. DISCUSSION

A. Accuracy and Applications

All the Bland-Altman plots demonstrate strong correlations between the two measurement methods. The near-zero MoD values indicate an absence of systematic bias, while the narrow LoA with values less than ± 0.5 s and ± 0.45 bpm reflect great agreement for 95% of the respiratory cycles. The calculated linear regression shows a slight proportional bias, with the worst being 0.93 for T_i of male volunteers. The r and p values indicate a strong linear relationship between the two measurement methods. Finally, the mean errors are less than 0.14 s, with the median being less than 0.09 s, showing a great agreement between the sensor and the reference.

The results show a good standing against state-of-the-art respiratory monitoring methods summarized in Table V. Compared to Near-field Coherent Sensing (NCS) method reported in [22], which requires two separate TX and RX antennas for operation, SABioS demonstrates a significantly lower LoA by a factor of 18 in estimating BR through a single antenna. The reported MAPE for flexible fiber [11], FBG [13], and temperature facemask [18] methods in estimating BR also show a less accuracy in these works compared to SABioS. A similar trend can also be observed against WaveFlex sensor [12], piezoresistive sensor [15], RIP [16] and radar-based [19] works.

The accuracy of the RFID-based work in detecting BR is dependent on the distance and angle between the antenna sensor and the tag reader station, which shows a MAPE of $100\% - 94.21\% = 5.79\%$ for reader placed 1.2 m away from the tag [20]. Similarly, the Bluetooth RSSI method requires users to stay within close proximity of a host station for wearable devices to connect. Signal quality and dynamic range in this method can be significantly affected by path loss and external interference. The choice of frequency in this method is confined to the 2.4 GHz frequency band, therefore limiting the sensitivity of antenna sensors to PSE. Moreover, the bit

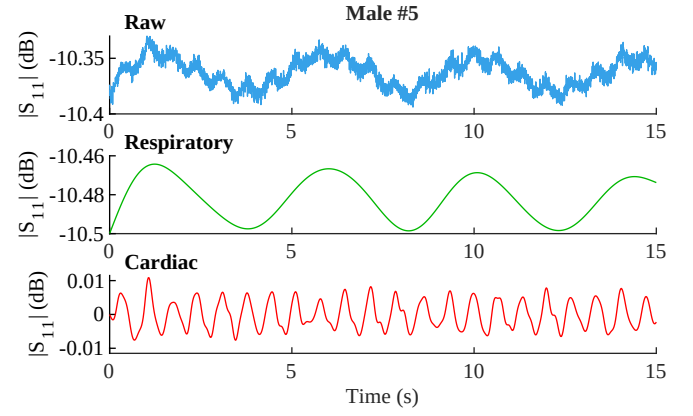


Fig. 9. The sample raw signal with respiratory and cardiac activity signals extracted.

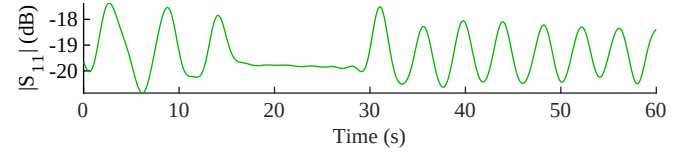


Fig. 10. A simulated example of Sleep Apnea.

resolution and sampling rate of the RSSI readout is usually low and pre-configured by the Bluetooth chip vendor, increasing the quantization noise. SABioS excels in accuracy for all the respiratory parameters compared to the previous RSSI-based work, both in MoD and LoA [21].

The single antenna design of SABioS and its independence from external devices during measurement significantly reduce interference, allowing for true mobility and continuous vital sign monitoring. Unlike methods primarily dependent on strain, such as RIP and FBG-based sensors, which necessitate a snug fit around the chest for respiratory monitoring, the PSE contributor allows the antenna sensor to operate effectively even when loosely placed on the chest, or over thick clothing.

The data recorded during this study suggest additional potential for measuring cardiac activity using the SABioS method due to PSE. Fig. 9 presents an example of the raw signal along with the extracted respiratory and cardiac activity signals. The cardiac signal is isolated by applying a high-pass FIR filter with a cutoff frequency of 1 Hz, corresponding to 60 beats per minute or higher. However, the accuracy of this cardiac activity signal remains to be confirmed. A future study incorporating a refined experimental protocol and a reference device for monitoring cardiac activity could provide more definitive evidence regarding the accuracy and reliability of SABioS for cardiac monitoring.

The high temporal accuracy of SABioS, particularly in detecting breathing rate (BR), suggests the potential to assess conditions such as hyperventilation and hypoventilation, which are associated with abnormally high or low BR, respectively. Additionally, this accuracy may offer insights into detecting sleep apnea events. Fig. 10 illustrates a volunteer simulating an apneic event by holding their breath, demonstrating the system's ability to capture pauses in the breathing cycle. This observation indicates that SABioS could potentially serve as a

tool for assessing apnea events. However, it's important to note that this simulation does not fully replicate the complexities of sleep apnea, as true apneic events involve subtle patterns in both respiration and heart rate during sleep [37]. Further studies involving actual sleep recordings are essential to validate the system's effectiveness in monitoring sleep apnea under real-world conditions.

B. Antennas, Regulations, and Safety

Fig. 11 shows sample signals recorded via various types of antennas to show the flexibility of choosing the antenna type for SABioS. The tested antennas include a PCB trace monopole antenna (KYOCERA AVX 1002089, Kyoto, Japan), the sinusoidal dipole antenna utilized in the current study [29], a ceramic chip antenna (Abracon ACAG1204-915-T, Spicewood, TX), and two custom-designed circular patch antennas with radii of 41.6 mm and 15.1 mm. It is evident that all of these antennas are capable of capturing respiratory activity, with a decreasing Signal-to-Noise Ratio (SNR) as frequency increases, confirming the discussions of Section II on choice of frequency for optimizing PSE. It can also be seen that the visibility of cardiac activity signal diminishes as frequency increases, which is expected according to previous discussions.

A key concern for on-body antenna applications is the Specific Absorption Rate (SAR) in tissues. Each type of antenna must be evaluated to ensure compliance with SAR limitations set by governing authorities. The sinusoidal dipole antenna employed in this study has already undergone SAR assessment to ensure biological safety [29], operating at a low power of -10 dBm results in a SAR of 1.38×10^{-3} W/Kg, which is more than three orders of magnitude below the maximum 1.6 W/Kg allowed in North America. The SABioS

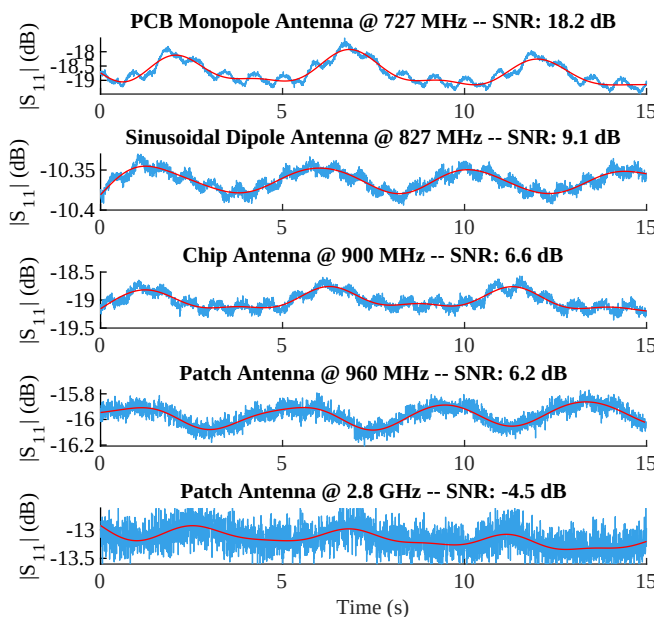


Fig. 11. Raw signal recorded via different antennas. Red trace shows the filtered BR signal. All examples are rigid antennas and solely benefiting from PSE, except the sinusoidal dipole antenna.

method does not require a high power to be fed to the antenna, and can measure vital signs as long as the detector is sufficiently sensitive. We have successfully tested power levels as low as -50 dBm in the laboratory environment for respiratory monitoring.

C. Robustness and Reliability

A set of preliminary results covering different configurations and basic activities are provided to demonstrate the robustness of SABioS. Fig. 12 display the samples recorded with the antenna sensor positioned on the thoracic back and abdomen areas, highlighting SABioS's ability to measure respiratory activity in both locations, with the measurement from the abdomen area attributed to ADE. However, according to our observations, unlike placement on the chest, the signal component associated with cardiac activity in both the thoracic back and abdomen placements becomes very weak and cannot be reliably extracted, as the sensor is positioned too far from the heart. This distance limits the ability of PSE to effectively detect cardiac activity in such configurations.

Additionally, Fig. 13a presents sample signals corresponding to different breathing profiles of shallow, normal, and deep breathing, demonstrating SABioS's capability to detect variations in respiratory amplitude. The signal shown in Fig. 13b corresponds to a person speaking, clearly indicating moments of inhalation along with gradual slopes that represent exhalation during speech. Fig. 13c illustrates a sample signal during head movements, which evidently has no significant effect on the recorded signal. Furthermore, Fig. 13d presents the case of moving hands while working at a table, which can cause slight variations in the baseline of the respiratory signal due to chest muscle contractions following hand movements.

Finally, Fig. 13e presents a sample signal recorded while the volunteer was walking. The volunteer was instructed to walk at a normal speed along a circular path due to the limited length of the wired connections to the VNA and the PC. Since the frequency of steps during walking is typically higher than the respiratory frequency, ranging from 60 to 119 steps per minute [38], a basic approach to reduce artifacts introduced by walking in the respiratory activity signal is using a simple low-pass filtering. However, further studies are needed to investigate the various movement artifacts in

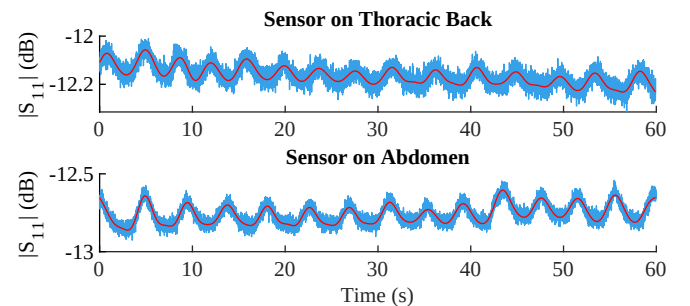


Fig. 12. Sample raw signals recorded while sensor placed on the thoracic back and the abdomen regions. Red trace shows the filtered BR signal. All examples are recorded using the sinusoidal dipole antenna.

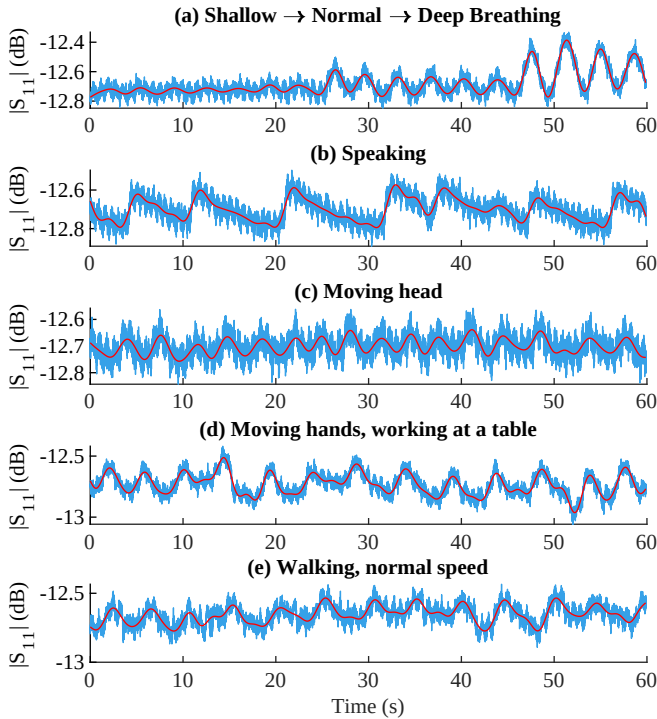


Fig. 13. Sample raw signals recorded during different activities. Red trace shows the filtered BR signal. All examples are recorded using the sinusoidal dipole antenna.

greater depth and to develop adaptive methods to minimize body movement effects.

The measurements obtained via SABioS are based on detecting relative changes in the raw signal. Adjustments in how the antenna is worn may shift the baseline of the acquired signal. While the adjustments are being made, artifacts may temporarily disrupt the signal, but once the adjustments stop, a new baseline can be established, allowing vital signs to be monitored again. The effects of wearing layers over the antenna sensor are discussed in Supplementary Material Section A, which examines the baseline shifts during transition periods when the antenna's configuration changes, such as when a cross-shoulder strap is placed over the sensor or extra layers like a sweater, raincoat, or hoodie with a metallic zipper are added on top. While these changes cause minor baseline shifts, the signal remains detectable after the brief disruption during the transition.

VI. CONCLUSION

The single antenna bio-sensing (SABioS) method, capable of monitoring respiratory activity and potentially extracting cardiac activity, was presented. The operation principle, contributing factors, and requirements for achieving optimal sensing were theoretically elaborated and validated through simulations. A preliminary experimental setup was established, recording signals from 14 volunteers using both the SABioS system and a medical-grade reference device. The sensor's accuracy and agreement with the reference device were analyzed using Bland-Altman analysis, linear regression, and various error metrics.

SABioS operates with various antenna types, providing design flexibility and application adaptability. It does not require a tight fit around the chest, direct skin contact, or any nearby external devices for operation, allowing for a mobile, comfortable, and non-invasive implementation of this biosensing method for continuous vital sign monitoring. As discussed, this method can also potentially be used to monitor the subject's cardiac activity using the same single sensor. It was also demonstrated that this method also has the capability to detect breathing pauses, which may aid in identifying apneic events.

This work establishes a foundation for future research in vital sign monitoring via SABioS. Our initial goal moving forward is to validate this system further by expanding upon the preliminary results with a larger, more diverse population, including real patients in clinical settings. Additionally, we aim to thoroughly investigate the accuracy and reliability of cardiac monitoring using this system against a standard medical-grade reference device.

Looking ahead, we are exploring the implementation of this method in a miniaturized and portable system to enable real-time and continuous monitoring, which will be invaluable for long-term patient care, sleep monitoring, and sports tracking applications. Furthermore, an experimental comparison between different antenna types is essential to assess their effectiveness and accuracy in acquiring both respiratory and cardiac activity. This evaluation will also examine their robustness against artifacts, clothing, and other environmental variables, ultimately aiming to optimize SABioS for the best possible performance.

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